

A Lieb–Robinson bound for the Kogut–Susskind Hamiltonian, with velocity $\propto Nc/g^2$

The Hamiltonian of the obligations map has exactly the structure for which a Lieb–Robinson bound survives an unbounded generator: its unbounded part is a sum of commuting single-link terms, and its interaction is a bounded finite-range sum of plaquette terms. Passing to the interaction picture generated by the electric term leaves a time-dependent bounded interaction of unchanged norm and range, so the standard theorem applies and gives, for observables A on a set X of links and B on Y ,

$$\|[A(t), B]\| \leq 2\|A\| \|B\| |X| e^{-(d(X,Y)-v|t|)/a}, \quad v = 2e a J/\hbar = 32e \frac{\dim(G)_f c}{g^2},$$

where $J = \max_\ell \sum_{p \ni \ell} \|w_p\| = 16N\hbar c/(g^2 a)$ for $SU(N)$ in the fundamental normalization, so $v \leq C Nc/g^2$ with C a pure number, **uniform in the lattice size and in the volume**. Three consequences. Combined with T2 and the Hastings–Koma and Nachtergaele–Sims clustering theorems, the strong-coupling phase has exponential decay of correlations with correlation length $\xi \leq v\hbar/\Delta \simeq C' a N/(\gamma C_2 g^4)$, shrinking as g^4 grows. The velocity diverges as $g \rightarrow 0$, so the lattice light cone opens faster than c at weak coupling, and the recovery of relativistic causality is one more statement that holds only along the scaling curve. And the bound supplies clustering while leaving the openness of the gapped set of the critical-coupling note open, because the available stability theorems additionally require frustration-freeness, which the Kogut–Susskind Hamiltonian lacks. The reduction to the standard theorem is the content here; the standard theorem is cited. Constants explicit; nothing promoted.

1. The structure that makes the unbounded term harmless

Take the links as the sites of the lattice and write, as in the obligations map §1,

$$H = H_E + W, \quad H_E = \frac{\hbar c g^2}{2a} \sum_{\ell \in \mathcal{E}} (-\Delta_\ell), \quad W = \sum_{p \in \mathcal{P}} w_p, \quad w_p = \frac{2\hbar c}{a g^2} (N - \text{Re tr } U_p).$$

Three properties are all that the argument uses.

1. *The unbounded part is single-site and commuting.* $-\Delta_\ell$ acts on the single factor $L^2(G)$ of link ℓ , and the terms for different links commute. Each generates a one-parameter unitary group $e^{-iscg^2\Delta_\ell/(2a)}$ whose adjoint action preserves the algebra of that link.
2. *The interaction is bounded.* $0 \leq w_p \leq \frac{4N\hbar c}{ag^2}$ in operator norm for $SU(N)$, since $|\text{Re tr } U| \leq N$; for a general compact G in a faithful representation of dimension $\dim_f$ replace N by $\dim_f$.
3. *The range is one plaquette.* Each w_p acts on the four links of p ; a given link lies in exactly four plaquettes in three spatial dimensions (two planes containing its direction, two plaquettes on either side in each). Hence

$$J := \max_\ell \sum_{p \ni \ell} \|w_p\| \leq 4 \cdot \frac{4N\hbar c}{ag^2} = \frac{16N\hbar c}{ag^2}.$$

2. The reduction and the bound

Proposition 1. Let $\tau_t(A) = e^{itH/\hbar} A e^{-itH/\hbar}$. For A supported on a set X of links and B on a disjoint set Y , with $d(X, Y)$ the lattice distance,

$$\|[\tau_t(A), B]\| \leq 2\|A\| \|B\| |X| \exp\left[\frac{v|t| - d(X, Y)}{a}\right], \quad v = \frac{2e a J}{\hbar} \leq \frac{32e Nc}{g^2}.$$

Proof. Let $U_E(t) = e^{-itH_E/\hbar}$ and define the interaction-picture interaction $\tilde{w}_p(t) = U_E(t)^* w_p U_E(t)$. Since H_E is a sum of commuting single-link terms, $U_E(t)$ factorizes over links and its adjoint action preserves each link algebra, so $\tilde{w}_p(t)$ is supported on the same four links and $\|\tilde{w}_p(t)\| = \|w_p\|$ for all t . The full evolution factorizes as $e^{-itH/\hbar} = U_E(t) \tilde{U}(t)$ with $\tilde{U}$ generated by the time-dependent bounded interaction $\tilde{W}(t) = \sum_p \tilde{w}_p(t)$, and conjugation by $U_E(t)$ preserves supports and norms. The Lieb–Robinson theorem for time-dependent bounded finite-range interactions (Lieb–Robinson, Commun. Math. Phys. 28 (1972) 251; Nachtergaele–Sims, Commun. Math. Phys. 265 (2006) 119, in the form used for the clustering theorem; metadata level) applies to $\tilde{W}$ with the constant J of Section 1, and transferring back through U_E gives the statement with the same constants. $\square$

The same argument is the one used for oscillator lattices, where the unbounded on-site harmonic part is treated exactly and the bounded anharmonic interaction carries the velocity (Nachtergaele–Raz–Schlein–Sims, Commun. Math. Phys. 286 (2008) 1073; metadata level). The gauge constraint plays no role: the bound holds on the full Hilbert space and restricts to the physical subspace, which is invariant under H .

Remark (what fails without the structure). If the electric term coupled two links, its unboundedness would enter the iteration and no bound of this form would follow. The single-link form of $\sum_a E_\ell^a E_\ell^a$ is therefore load-bearing, and it is the same feature that made the strong-coupling hypothesis check of the T2 note work.

3. Correlation length at strong coupling

Corollary 2. For $g \geq g_0(G)$, the infinite-volume ground state of the Kogut–Susskind Hamiltonian has exponentially decaying connected correlations of local gauge-invariant observables, with correlation length

$$\xi \leq \frac{C v \hbar}{\Delta} \leq \frac{C' a N}{\gamma C_2(R_{\min}) g^4},$$

for pure numbers C, C' .

Proof. Proposition 1 supplies the Lieb–Robinson bound; the T2 note supplies the uniform gap $\Delta \geq \gamma(g^2/2)C_2\hbar c/a$ and a unique ground state. The exponential clustering theorem for gapped local Hamiltonians (Nachtergaele–Sims, Commun. Math. Phys. 265 (2006) 119; Hastings–Koma, Commun. Math. Phys. 265 (2006) 781; both metadata level) gives decay with length of order $v\hbar/\Delta$. Substituting $v \leq 32eNc/g^2$ and $\Delta \geq \gamma C_2 g^2 \hbar c/(2a)$ gives the displayed form. $\square$

The correlation length shrinks like g^{-4} as the coupling grows, which matches the strong-coupling expansion, where the leading correlation of two plaquettes at distance d costs a tube of $\sim d/a$ plaquettes each suppressed by g^{-4} . Yarotsky’s conclusion 3, used in the T2 note, already asserts exponential clustering; Corollary 2 adds the explicit length in terms of the gap and the velocity.

4. The velocity at weak coupling, and lattice causality

The velocity $v \leq 32eNc/g^2$ exceeds c once $g^2 < 32eN$, that is throughout the weak-coupling regime relevant to the continuum limit. The lattice theory has a light cone set by the interaction strength rather than by the Minkowski metric, and the two agree only after the scaling limit restores relativistic causality. This is one more obligation in the same family as T3 and T4: the axioms require exact locality of the continuum theory, and here the lattice bound alone degrades as $g \rightarrow 0$. It also gives a second reading of the crossover of the valley note: the lattice light cone crosses a box of side L in time $L/v \sim Lg^2/(Nc)$, shorter than the inverse zero-mode gap $L/(g^{2/3}c)$ whenever $g^{4/3} < 1/N$, so at weak coupling the whole box is causally connected on the timescale of the gap, and the small-volume description is consistent.

5. What the bound supplies, and what openness still needs

Lieb–Robinson bounds enter gap-stability theorems as one hypothesis among several. The available theorems for perturbations of gapped local Hamiltonians, including the topological-order stability results used in the T2 note, additionally require the unperturbed Hamiltonian to be frustration-free with a locally topologically ordered ground space. The Kogut–Susskind Hamiltonian is frustration-free only at $g = \infty$, where the ground state is the constant function annihilated by every $-\Delta_\ell$; at finite g the electric and magnetic terms compete and no state minimizes both. So openness of the gapped set needs either a stability theorem for non-frustration-free Hamiltonians with a uniform gap, or an argument specific to gauge theory. Proposition 1 removes one of the two obstacles that stood in the way, and names the remaining one precisely.

6. Consequence for STATE

The Lieb–Robinson question raised in the critical-coupling note §5 is settled: the bound holds with velocity $32eNc/g^2$ uniformly in volume, because the unbounded part of the Hamiltonian is single-link and commuting. Openness of the gapped set now rests on a single missing ingredient, a stability theorem that does not assume frustration-freeness. Two further items follow from the bound itself: the explicit strong-coupling correlation length of Corollary 2, and the observation that the lattice light cone opens faster than c at weak coupling, which belongs with T3 and T4 as a statement that holds only along the scaling curve.